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Core Web Design Principles

Principles of Website Design

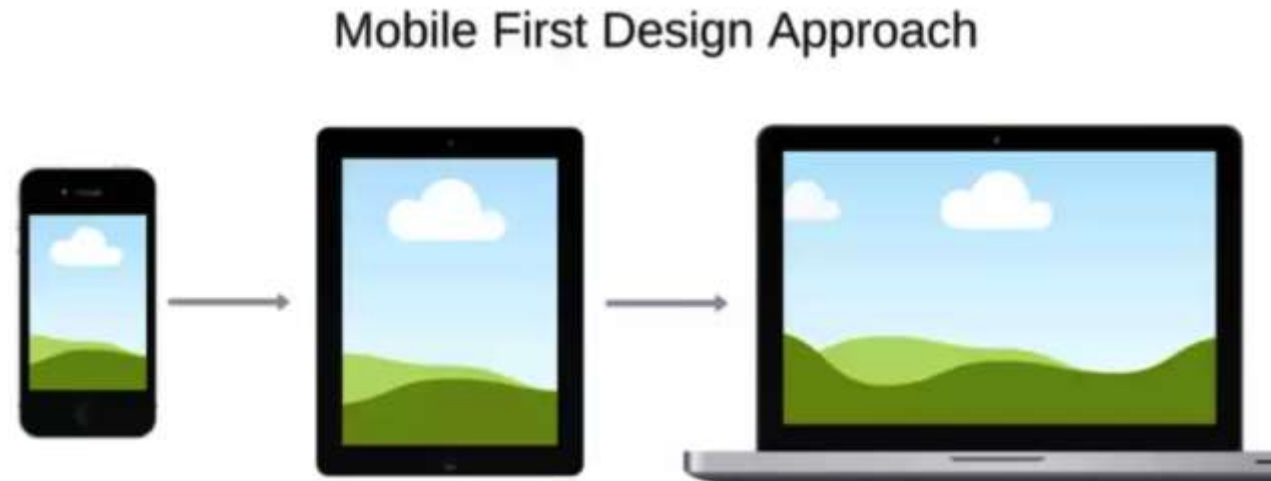
To ensure that websites are functional, user-friendly, and effective, website development is guided by various principles and best practices.

1. User-Centric Design - Understand their needs, preferences, and behaviors to create a website that meets their expectations.
2. Responsive Design – design should be responsive and adaptable to different screen sizes and devices, including mobile phones, tablets, and desktops.



Principles of Website Design

3. Mobile-First Approach - start with mobile design and progressively enhance for larger screens.



Principles of Website Design

4. Usability and User Experience (UX) – interface should be user-friendly to keep the users/visitors engaged with your website by ensuring easy navigation, readable content, etc
5. Performance Optimization - minimizing load times and ensuring fast page loading
 - A slow site drives users away.
 - Compress images, minimize scripts, and reduce unnecessary assets.

Four Elements of User Experience

Value Is it useful?	Usability Is it easy to use?
Adoptability Is it easy to start using?	Desirability Is it fun and engaging?

Principles of Website Design

6. Content Quality - provide high-quality, relevant, and up-to-date content that meets the needs of your target audience. Content should be well-organized and easy to understand.
7. Search Engine Optimization (SEO) - implement on-page and technical SEO best practices to improve visibility in search engine results pages (SERPs). This includes keyword research, meta tags, and quality backlinks
8. Security – implement security measures such as SSL encryption, regular software updates, firewalls, and strong authentication for administrators.

Principles of Website Design

9. Scalability and Flexibility – design your site to allow growth and expansion. Use flexible coding and modular components that can adapt to changing requirements.
10. Cross-Browser Compatibility – ensure that website works correctly and looks consistently across different web browsers
11. Accessibility - Make the website accessible to all users
12. Testing and Quality Assurance -- Conduct thorough testing to identify and fix issues related to functionality, performance, compatibility, and security before launching the website.

Principles of Website Design

- 13. Feedback and Iteration -- Collect feedback from users and stakeholders to make continuous improvements to the website.
- 14. Legal and Privacy Compliance - Ensure that it complies with relevant laws and regulations, including privacy policies, terms of service, and copyright laws.
- 15. Backup and Disaster Recovery
- 16. Documentation and Maintenance -- Maintain detailed documentation of the website's architecture, code, and configurations.

Web Standards (W3C) and Web Design

Standards	How It Relates to Web Design Principles
1. Security	• Influences design of user interfaces (e.g., secure login forms, password visibility toggles) • Impacts trust and credibility of visual design (e.g., security badges, SSL indicators) • Requires clear feedback design for errors and warnings (e.g., invalid password alerts)
2. Internationalization	• Affects typography, layout, and content structure to accommodate different languages, scripts, and reading directions (LTR vs RTL) • Requires flexible design to support locale-specific formatting (dates, currencies) • Encourages designers to avoid hardcoding text into graphics

Web Standards (W3C) and Web Design

Standards	How It Relates to Web Design Principles
3. Privacy	• Informs the placement and design of consent dialogs (e.g., cookie banners, opt-in forms) • Encourages transparent UX that explains data usage • Requires careful form design to collect only essential data
4. Accessibility	• Directly overlaps with key design principles like contrast, typography, navigation, and visual hierarchy • Demands semantic structure, keyboard-friendly navigation, and alt text • Forces designers to consider non-visual users (screen readers, voice control)

Planning Your Website Design

"Failure to plan is planning to fail." - Benjamin Franklin

1. What? Define Your Website's Purpose and Goals

- define the purpose of your website.
- Determine what you want to achieve with it,
 - whether it's to provide information,
 - sell products, generate leads, or
 - build an online community.

2. Who? Identify Your Target Audience

- Know your audience
- Understand their demographics, needs, preferences, and pain points.
- Customize your design to cater to the interests and behaviors of your target audience.

Planning Your Website Design

3. Conduct Market Research/ Conduct an environment scanning
 - Research to identify trends, best practices, and areas where you can differentiate your site.
 - Analyze what works well for competitors and what you can improve upon.
4. Choose Your Domain Name and Hosting
 - Select a domain name (upmin.edu.ph) that is relevant to your brand and easy to remember.
5. Create a Sitemap -- outlines the main sections and pages of your website, organize content hierarchically to guide users effectively.

Planning Your Website Design

6. Wireframing and Prototyping -- Create wireframes or prototypes to visualize the layout and functionality of your website. These sketches help you plan the design before actual development begins.
7. Content Strategy - Plan your content strategy, including the type of content (text, images, videos), tone, voice, and messaging.
8. Design Style and Branding, Responsive Design, Accessibility and Visibility (SEO)
9. Development and Testing -- Conduct thorough testing of your website's functionality, performance, compatibility, and security before launch.
10. Launch and Promotion -- launch it and promote it through various channels, such as social media, email marketing, and SEO optimization.

Planning Your Website Design

11. Monitoring and Maintenance

- Continuously monitor your website's performance, user feedback, and analytics.
- update content, plugins, and security features to ensure a seamless user experience.

12. Feedback and Iteration -- Collect feedback from users and stakeholders and use it to make improvements and updates to your website over time.

13. Legal and Privacy Compliance -- Ensure that your website complies with relevant laws and regulations, including privacy policies, terms of service, and copyright laws.



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Website Legal and Privacy Compliance

GDPR – General Data Protection Regulation

- GDPR is a European Union (EU) regulation that applies to the processing of personal data of EU residents, regardless of where the processing takes place.
- GDPR protects the rights and privacy of individuals, known as data subjects (individuals), whose personal data is processed.
- Data subjects must provide clear and informed consent for their data to be processed.
- GDPR grants data subjects rights, including the right to access, rectify, erase, and restrict the processing of their data.
- Organizations must conduct DPIAs for high-risk data processing activities to assess and mitigate potential privacy risks.

Website Legal and Privacy Compliance

GDPR – General Data Protection Regulation

Data Breach Notification - Organizations must report data breaches to supervisory authorities and notify affected data subjects within specified timeframes.

Penalties:

For less severe infringements, organizations may face fines of up to €10 million or 2% of their global annual turnover, whichever is higher.

For especially severe violations, listed in Art. 83(5) GDPR, the fine framework can be up to € 20 million, or 4 % of their total global turnover of the preceding fiscal year, whichever is higher.

References:

<https://www.onlineinnovations.com/article/the-difference-between-non-responsive-mobile-and-responsive-websites>

<https://www.altexsoft.com/blog/business/how-to-be-mobile-first-company/>